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What Is Reality?

in which spacetime is viewed as
curved on Sundays and flat on Mondays,
and horizons are made from
vacuum on Sundays and charge on Mondays,
but Sunday's experiments and Monday's experiments
agree in all details

Is spacetime *really* curved? Isn't it conceivable that spacetime is actually flat, but the clocks and rulers with which we measure it, and which we regard as *peifect* in the sense of Box 11.1, are actually rubbery? Might not even the most perfect of clocks slow down or speed up, and the most perfect of rulers shrink or expand, as we move them from point to point and change their orientations? Wouldn't such distortions of our clocks and rulers make a truly flat spacetime appear to be curved?

Yes.

Figure 11.1 gives a concrete example: the measurement of circumferences and radii around a nonspinning black hole. On the left is shown an embedding diagram for the hole's curved space. The space is curved in this diagram because we have chosen to define distances as though our rulers were *not* rubbery, as though they always hold their lengths fixed no matter where we place them and how we orient them. The rulers show the hole's horizon to have a circumference of 100 kilometers A circle of twice this circumference, 200 kilometers, is drawn around the hole, and the radial distance from the horizon to that circle is measured with a perfect ruler; the result is 37 kilometers. If space were flat, that radial distance